

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Semestertoets 1
VRAESTEL

Kopiereg voorbehou

Module EBN111
Elektrisiteit en Elektronika
Maart 2017

Department of Electrical, Electronic and
Computer Engineering

Semester Test 1
QUESTION PAPER

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Module EBN111
Electricity and Electronics
March 2017



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	46	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 min + 10 min to complete OCR form <i>90 min + 10 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		15 + 2 Answer sheets	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD en op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
All questions must be answered on the ANSWER SHEET and on the Optical Character Recognition (OCR) form.
- Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye

STEPS TO FOLLOW

Do steps 1 to 3 in the first 90 minutes. Do step 4 within the following 10 minutes.

STEP 1:

Do your own calculations on the QUESTION PAPER. The question paper is not taken in at the end of the test.

STEP 2:

Write down the answers for each question on the ANSWER SHEET. Be sure to include the units.

STEP 3:

Copy the answers from the ANSWER SHEET to the PRACTICE PAGE. Each block may only contain a single character. See example below:

Examples

01	–	1	.	4	5					m	A						
02	3	4	0							v							
03	1	5	.	2													
04	-	1								Ω							

The correct units for each question are shown on the PRACTICE PAGE.

STEP 4:

After 90 minutes carefully copy the answers from the PRACTICE PAGE to the green Optical Character Recognition (OCR) form. You have 10 minutes do this.

- The Assessment ID is: **2017EBN111S1**
- Use a mechanical pencil to complete the form.
- Each cell should only have one character.
- Characters may not touch or cross the cell.

INSTRUKSIES

Doen stappe 1 tot 3 in die eerste 90 minute. Doen stap 4 binne die daaropvolgende 10 minute.

STAP 1:

Doen jou eie berekeninge op die VRAESTEL. Die vraestel word nie aan die einde van die toets ingeneem nie.

STAP 2:

Skryf die antwoorde vir elke vraag op die ANTWOORDBLAD. Maak seker om eenhede in te sluit.

STAP 3:

Dra die antwoorde vanaf die ANTWOORDBLAD oor na die OEFENBLAD. Elke blokkie mag slegs een karakter bevat. Sien die voorbeeld hieronder.

Voorbeelde

01	–	1	.	4	5					m	A						
02	3	4	0							v							
03	1	5	.	2													
04	-	1								Ω							

Die korrekte eenhede vir al die vrae in die toets word reeds op die OEFENBLAD aangedui.

STAP 4:

Na 90 minute, dra die antwoorde van die OEFENBLAD noukeurig oor na die groen Optiese Karakter Herkenning (OKH) vorm. Jy het 10 minute om dit te doen.

- Die “Assessment ID” is **2017EBN111S1**
- Gebruik ‘n meganiese potlood om antwoorde in te vul.
- Elke blokkie mag slegs een karakter bevat.
- Karakters mag nie aan die grense van die blokkies raak of oor dit steek nie.

SECTION / AFDELING 1 [18]

Question 1 & 2 [2]		Vraag 1 & 2 [2]	
01	An electric stove with twee electric plates and an oven is used in preparing a meal as follows: • Plate 1 (1.8 kW) : 20 minutes, • Plate 2 (1.2 kW): 40 minutes, • Oven (3 kW): 30 minutes Calculate the cost of electricity used in preparing the meal. (Electricity costs R 2 per kWh. Give your answer in Rands.) [1]	01	‘n Elektriese stoof met twee elektriese plate en een oond word as volg gebruik om ‘n maaltyd voor te berei: • Plaat 1 (1.8 kW): 20 minute • Plaat 2 (1.2 kW): 40 minute • Oond (3 kW): 30 minute Wat is die elektrisiteitskoste om die maaltyd voor te berei? (Elektrisiteit kos R 2 per kWh. Gee die antwoorde in Rande.) [1]
02	If a constant 100 V is applied to an element, and the element uses 150 kJ over 1 min, what is the size of the current flowing through the element? [1]	02	Indien ‘n konstante 100 V oor ‘n element aangelê word, en die element gebruik 150 kJ oor 1 min, wat is die grootte van die stroom deur die element? [1]

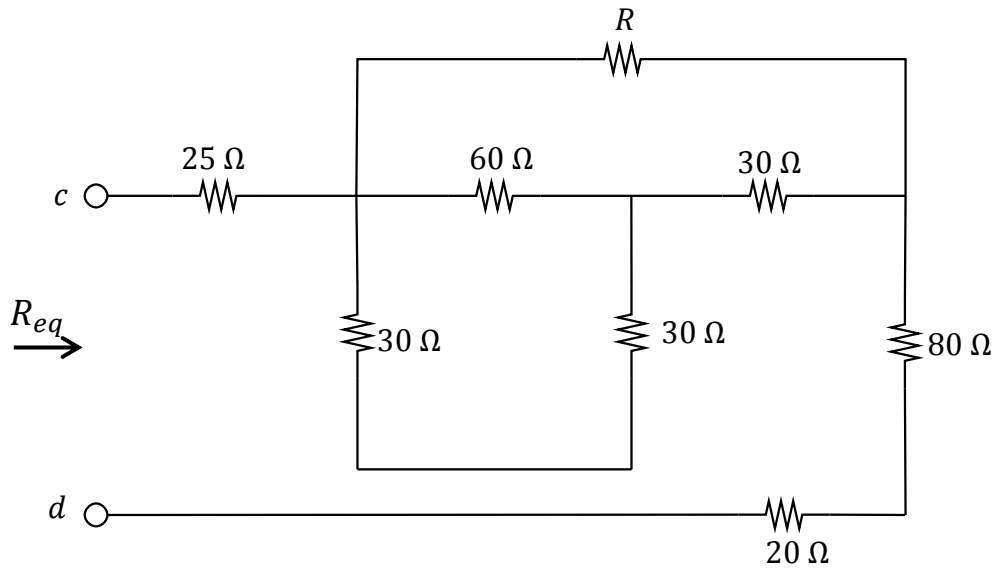
Question 3 & 4 [4]		Vraag 3 & 4 [4]	
Find the power related to following elements in the circuit below:		Bepaal die drywing geassosieer met die volgende elemente in onderstaande kringbaan:	
03	The independent voltage source.	03	Die onafhanklike spanningsbron.
04	The dependent current source.	04	Die afhanklike stroombron.

Question 5 [2]

Vraag 5 [2]

05 Find R if $R_{eq} = 161 \Omega$.

05 Bepaal R indien $R_{eq} = 161 \Omega$.

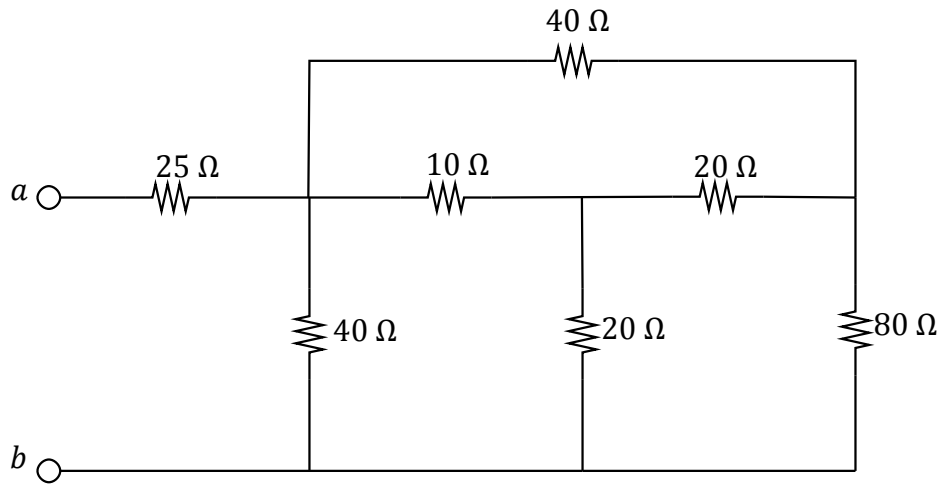


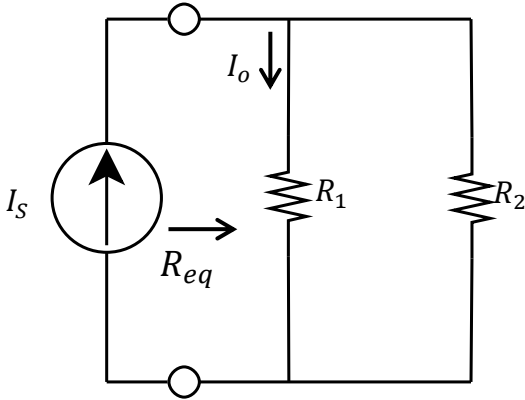
Question 6 [2]

Vraag 6 [2]

06 Determine R_{ab} .

06 Bepaal R_{ab} .

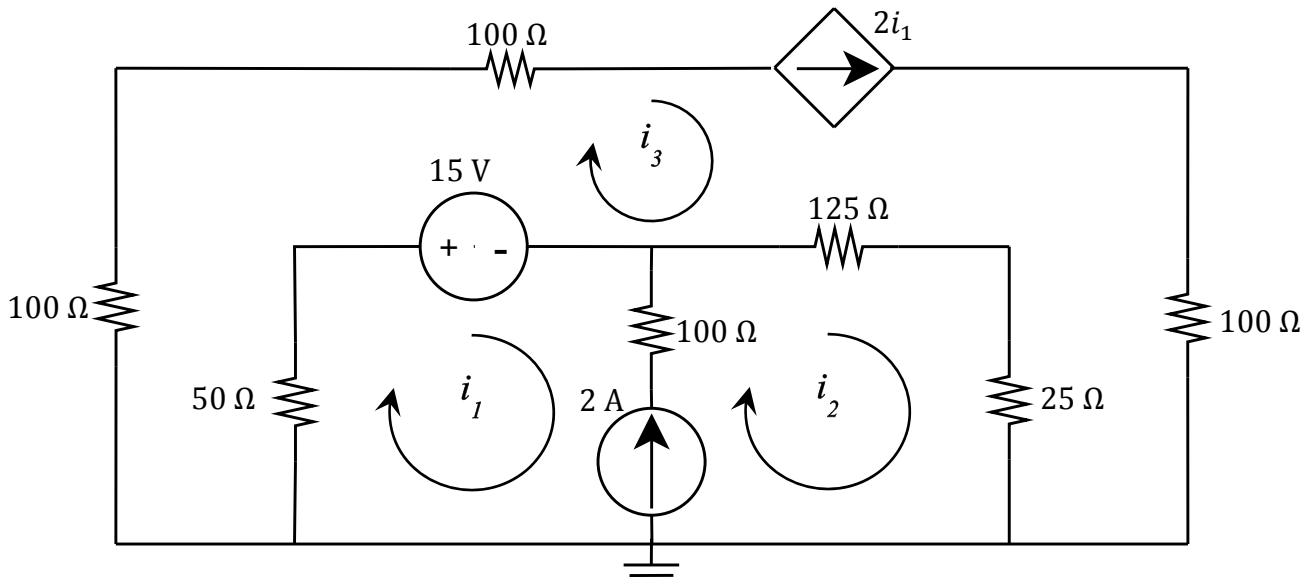


Question 7 & 8 [4]				Vraag 7 & 8 [4]			
For the circuit below it is given that $\frac{I_o}{I_S} = 0.2$ and $R_{eq} = 5 \text{ k}\Omega$. Find R_1 and R_2 .				Vir onderstaande kringbaan is dit gegee dat $\frac{I_o}{I_S} = 0.2$ en $R_{eq} = 5 \text{ k}\Omega$. Bepaal R_1 en R_2 .			
07	R_1	08	R_2	07	R_1	08	R_2
							

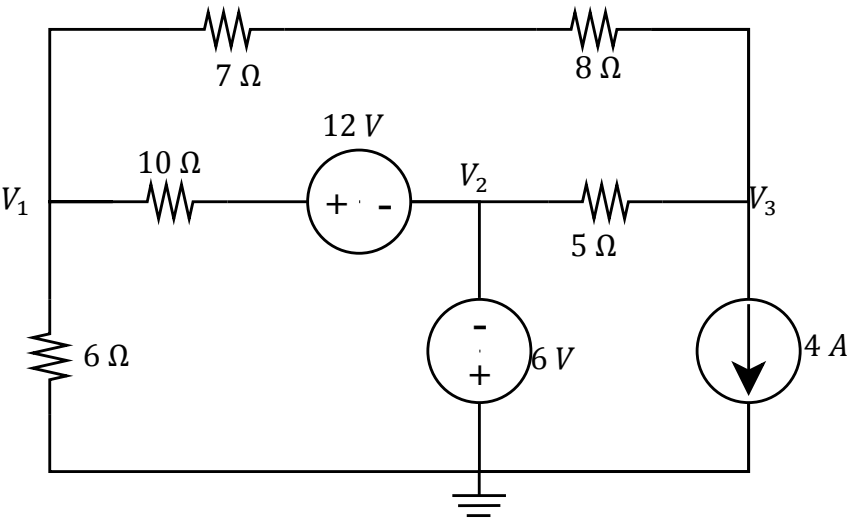
Question 9 & 10 [4]				Vraag 9 & 10 [4]			
If $I_0 = 5$ A, determine I_S and V_1 .				Indien $I_0 = 5$ A, bepaal I_S en V_1 .			
09	I_S	10	V_1	09	I_S	10	V_1

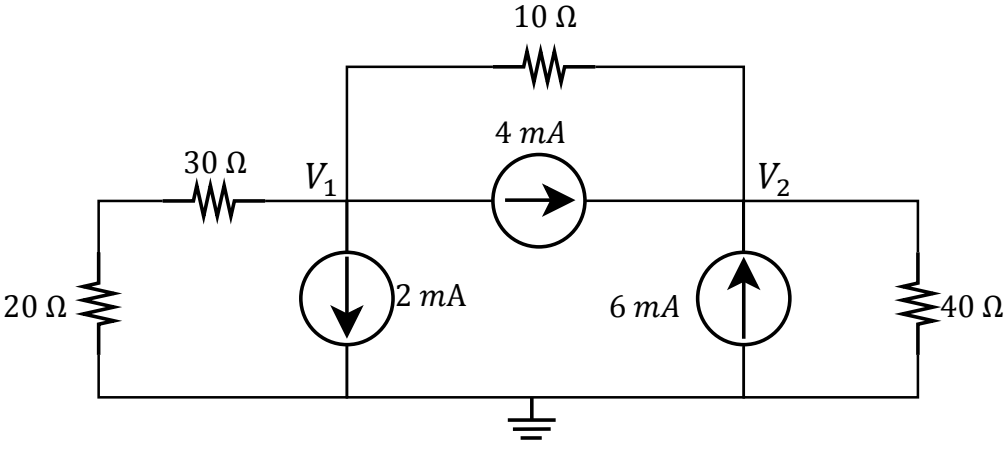
VRAAG/QUESTION 2 [20]

Question 11 & 12 [4]		Vraag 11 & 12 [4]	
11	Determine θ if $v_I = -2$ V. $\begin{bmatrix} \theta & -2 \\ 6 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \theta \\ -3\theta \end{bmatrix}$	11	Bepaal θ indien $v_I = -2$ V. $\begin{bmatrix} \theta & -2 \\ 6 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \theta \\ -3\theta \end{bmatrix}$
12	Determine σ if $i_2 = -2.6$ A. $\begin{bmatrix} 2 & 1 & 4 \\ 0 & \sigma & 1 \\ 2 & 4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$	12	Bepaal σ indien $i_2 = -2.6$ A. $\begin{bmatrix} 2 & 1 & 4 \\ 0 & \sigma & 1 \\ 2 & 4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$

Question 13-16 [4]			Vraag 13-16 [4]		
Use mesh analysis to set up two equations of the following form:			Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel:		
$ai_1 + bi_2 = -15$			$ai_1 + bi_2 = -15$		
$ci_1 + di_2 = 2$			$ci_1 + di_2 = 2$		
Calculate the unknown coefficients.			Bepaal die onbekende koëffisiënte.		
13	a	14	b	13	a
15	c	16	d	15	c
					

Question 17-20 [4]				Vraag 17-20 [4]			
Use nodal analysis to set up two equations of the following form: $eV_1 + fV_3 = 18$ $gV_1 + hV_3 = -78$ Calculate the unknown coefficients.				Gebruik nodeanalyse om twee vergelykings in die volgende vorm op te stel: $eV_1 + fV_3 = 18$ $gV_1 + hV_3 = -78$ Bepaal die onbekende koëffisiënte.			
17	e	18	f	17	e	18	f
19	g	20	h	19	g	20	h



Question 21-24 [2]				Vraag 21-24 [2]			
Use inspection to set up the matrix equation in terms of the node voltages.				Gebruik inspeksie om die matriksvergelykings in terme van die nodespannings op te stel.			
$\begin{bmatrix} k & l \\ -0.1 & p \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -6 \text{ mA} \\ q \end{bmatrix}$				$\begin{bmatrix} k & l \\ -0.1 & p \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -6 \text{ mA} \\ q \end{bmatrix}$			
21	k	22	l	21	k	22	l
23	p	24	q	23	p	24	q
							

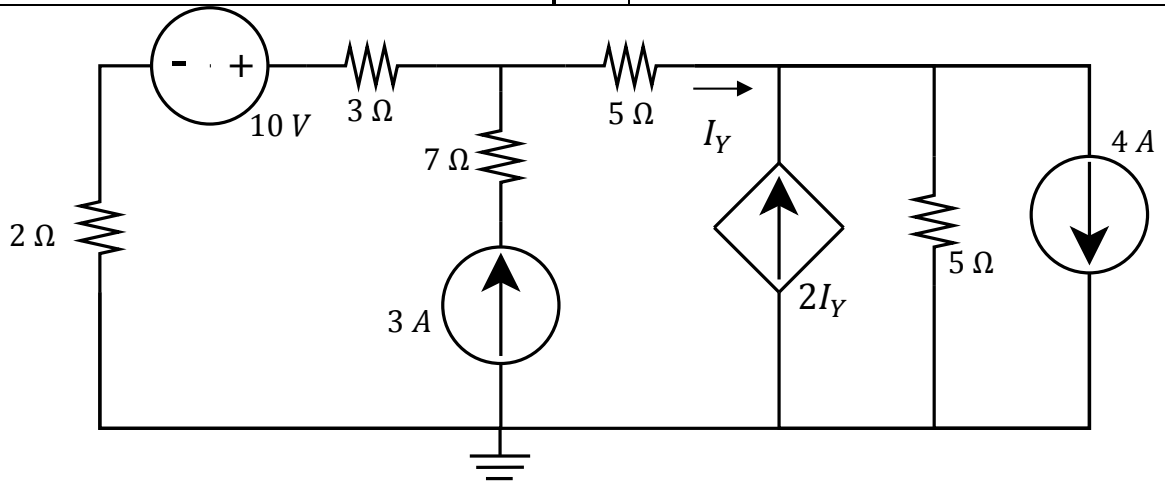
Question 25 [2]		Vraag 25 [2]	
25	Calculate V_0 using either nodal or mesh analysis.	25	Bepaal V_0 deur gebruik te maak van node- of lus-analise.

Question 26 & 27 [4]		Vraag 26 & 27 [4]	
It is given that $\beta = 99$ and $V_{BE} = 0.7V$.		Dit is gegee dat $\beta = 99$ en $V_{BE} = 0.7V$.	
26	Determine V_{CE} if $I_B = 25 \mu A$, $R_E = 0 \Omega$.	26	Determine V_{CE} if $I_B = 25 \mu A$, $R_E = 0 \Omega$.
27	Determine R_B if $I_B = 20 \mu A$, $V_S = 2V$, $R_E = 150 \Omega$.	27	Determine R_B if $I_B = 20 \mu A$, $V_S = 2V$, $R_E = 150 \Omega$.

$\beta = 99$
 $V_{BE} = 0.7 V$

VRAAG/QUESTION 3 [8]

Question 28 & 29 [4]		Vraag 28 & 29 [4]	
Using superposition and calculate:		Gebruik superposisie en bepaal:	
28	the current I_Y due to the 4 A current source alone.	28	die stroom I_Y slegs as gevolg van die 4 A stroombron.
29	the current I_Y due to the 10 V voltage source alone.	29	die stroom I_Y slegs as gevolg van die 10 V spanningsbron.



Question 30 & 31 [4]				Vraag 30 & 31 [4]			
Determine the Thevenin equivalent voltage V_{TH} and resistance R_{TH} w.r.t. terminals $a-b$.				Bepaal die Thevenin ekwivalente spanning V_{TH} en weerstand R_{TH} in terme van terminale $a-b$.			
30	V_{TH}	31	R_{TH}	30	V_{TH}	31	R_{TH}
